Dirichlet Integral

Senan Sekhon

Version of June 30, 2026, 07:15:12 (UTC+1)

0 Introduction

Lemma 0.1 (Dirichlet Kernel). For all $n \in \mathbb{N}_0$ and all $x \in \mathbb{R}$, we have:

$$1 + 2\sum_{k=1}^{n} \cos(2kx) = \frac{\sin((2n+1)x)}{\sin(x)}$$

Proof. Clearly the identity holds for $n = 0$ as both sides are 1. Suppose the identity holds for some $n \in \mathbb{N}_0$. We want to show it holds for $n+1$. We have:

$$\begin{aligned}1 + 2\sum_{k=1}^{n+1} \cos(2kx) &= 1 + 2\sum_{k=1}^{n} \cos(2kx) + 2\cos((2n+2)x) = \frac{\sin((2n+1)x)}{\sin(x)} + 2\cos((2n+2)x) \\ &= \frac{\sin((2n+1)x) + 2\sin(x)\cos((2n+2)x)}{\sin(x)}\end{aligned}$$

Note that $\sin((2n+3)x) - \sin((2n+1)x) = 2\sin(x)\cos((2n+2)x)$. This yields:

$$1 + 2\sum_{k=1}^{n+1} \cos(2kx) = \frac{\sin((2n+1)x) + \sin((2n+3)x) - \sin((2n+1)x)}{\sin(x)} = \frac{\sin((2n+3)x)}{\sin(x)}$$

Thus the identity holds for $n+1$, and so it holds for all $n \in \mathbb{N}_0$. 😎

Theorem 0.2 (Dirichlet Integral).

$$\int_0^\infty \frac{\sin(x)}{x}\,dx = \frac{\pi}{2}$$

Proof. Define $K : \left[0, \frac{\pi}{2}\right] \to \mathbb{R}$ as follows:

$$K(x) = \begin{cases} \dfrac{\sin((2n+1)x)}{\sin(x)} & x \neq 0 \\ 2n+1 & x = 0 \end{cases}$$

Clearly K is continuous on $\left(0, \frac{\pi}{2}\right]$, and:

$$\begin{aligned}\lim_{x\to 0^+} K(x) &= \lim_{x\to 0^+} \frac{\sin((2n+1)x)}{\sin(x)} \\ &= \lim_{x\to 0^+} (2n+1)\frac{\sin((2n+1)x)}{(2n+1)x}\,\frac{x}{\sin(x)} \\ &= (2n+1)\left(\lim_{x\to 0^+} \frac{\sin((2n+1)x)}{(2n+1)x}\right)\left(\lim_{x\to 0^+} \frac{x}{\sin(x)}\right) \\ &= (2n+1)\cdot 1 \cdot 1 \\ &= 2n+1\end{aligned}$$

Thus K is continuous (and thus Riemann integrable) on $\left[0, \frac{\pi}{2}\right]$. Using the Dirichlet kernel, we have:

$$\int_0^{\pi/2} \frac{\sin((2n+1)x)}{\sin(x)}\,dx = \int_0^{\pi/2} \left(1 + 2\sum_{k=1}^{n} \cos(2kx)\right) dx = \int_0^{\pi/2} 1\,dx + 2\sum_{k=1}^{n}\int_0^{\pi/2} \cos(2kx)\,dx = \frac{\pi}{2}$$

We also have:

$$\int_0^{\pi/2} \frac{\sin((2n+1)x)}{x}\,dx = \int_0^{(2n+1)\pi/2} \frac{\sin(x)}{x}\,dx$$

As $n \to \infty$, this tends to $\int_0^\infty \frac{\sin(x)}{x}\,dx$. Define $g : (0, \frac{\pi}{2}] \to \mathbb{R}$ by $g(x) = \frac{1}{x} - \frac{1}{\sin(x)}$. Clearly g is continuous on $\left(0, \frac{\pi}{2}\right]$.

Also, using L'Hôpital's rule twice, we have:

$$\lim_{x\to 0^+} g(x) = \lim_{x\to 0^+} \frac{\sin(x) - x}{x\sin(x)} = \lim_{x\to 0^+} \frac{\cos(x) - 1}{\sin(x) + x\cos(x)} = \lim_{x\to 0^+} \frac{-\sin(x)}{2\cos(x) - x\sin(x)} = \frac{0}{2-0} = 0$$

Thus g can be made continuous on $\left[0, \frac{\pi}{2}\right]$ by setting $g(0) = 0$. By the Riemann-Lebesgue lemma, we have $\int_0^{\pi/2} \left(\frac{1}{x} - \frac{1}{\sin(x)}\right) \sin((2n+1)x)\, dx \to 0$. 😎

Lemma 0.3. Suppose $f \in C[0,\pi]$. Define:

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin(nx)\, dx \qquad S_n = \int_0^{\pi} \left(f(x) - \sum_{k=1}^{n} b_k \sin(kx)\right)^2 dx$$

Then:

$$S_n = \int_0^{\pi} f(x)^2\, dx - \frac{2}{\pi}\sum_{k=1}^{n} b_k^2$$